DICTO: Collected Geometric Works

Papers, findings and open apps on polyhedra, space-filling and icosahedral order, by DICTO

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Licence CC BY-SA 4.0 · The apps: kaleidohedra.dictospheres.com, rhombiverse.dictospheres.com, polyhedraverse.dictospheres.com.
All works, findings and names are DICTO's. Computation and checking with Claude (Anthropic).

> **About.** DICTO is a geometric sculptor in Osaka. The work grows out of building with recycled PET bottles and from three open apps on one shared geometry: Kaleidohedra, Rhombiverse and Polyhedraverse.
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> This record gathers all of DICTO's published geometry in one place. It opens with a short essay, *Quanta of Space*, then lists the papers in four chapters: **Atoms** (new solids), **Bonds** (how pieces fill space), **Shells** (layers round a core) and **Cycles** (rules that turn one shell into the next).
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> Each paper has its own DICTO DOI, which always opens its newest version. The numbers (#1, #2 …) are the paper numbers on dictospheres.com; DICTO's findings are numbered separately, F1, F2 …. What changed in each version is listed at the end.

Introduction: Quanta of Space

In plain words. Is space smooth all the way down, or is it built from pieces, as a wall is built from stones? Thinkers have asked this for two and a half thousand years. These works take the question literally: they look for exact sets of solid pieces that fit together, with no gaps, into shells, cycles and whole spaces.

1. Atoms and the void

Leucippus and Democritus taught that the world is made of indivisible atoms moving in the void. The atoms differ only in shape, arrangement and position. Lucretius put the teaching into verse: “nothing exists per se except atoms and the void”. The atoms were pieces, but no one said what shapes they had.

2. Plato’s tile kit

Plato gave them shapes. In the *Timaeus* the four elements are four regular solids: fire is the tetrahedron, air the octahedron, water the icosahedron and earth the cube. All four faces are built from just two right triangles. The fifth solid, the dodecahedron, was used “for the whole”. It was the first tile kit for matter: a small set of exact pieces from which everything is assembled.

3. Aristotle’s mistake

Aristotle answered that solid atoms must fill space. He claimed (*On the Heavens* III.8) that among solids only two do: the pyramid and the cube. The cube does. The regular tetrahedron

does not, and the error stood for centuries. Paper #44, *Aristotle's Pyramid*, asks what he could have meant. It proves which face-pyramids of the Platonic solids can fill space.

4. Kepler's packings

Johannes Kepler nested the five solids between the planets' spheres (*Mysterium Cosmographicum*, 1596). In *The Six-Cornered Snowflake* (1611) he explained the snowflake by the packing of equal small units, the question that became the Kepler conjecture. Paper #8, *The Euclid–Kepler–Pacioli Cell*, finds all five Platonic solids nested on 13 points of one repeating cell.

5. The quanta of modern science

In 1900 Max Planck found that energy comes in quanta. In loop quantum gravity, space itself does too. Carlo Rovelli and Lee Smolin showed in 1995 that area and volume come in discrete amounts. In 2011 Eugenio Bianchi, Pietro Doná and Simone Speziale showed that each quantum of volume can be read as a small convex polyhedron, fixed by the areas and directions of its faces. Space becomes a network of polyhedra joined face to face.

There is an honest difference. In that theory two neighbouring quanta share a face of the same area, but not, in general, of the same shape (Freidel and Speziale, 2010). The pieces need not fit. Our pieces always fit exactly, corner to corner. Other approaches, such as causal sets, build spacetime from points rather than solids.

This collection makes no claim about physics. It shares the question and the form: what are the smallest exact pieces from which space can be built, and how do they join?

6. The works as a study of quanta

Read this way, the works fall into four chapters.

- **Atoms.** Single solids, each an exact new piece: the DICTO Ogdoecontahedron (OG), the Hecatodihedron, the Jewel, the Hexa, AXE, CLEO and FUJI, and the Reuleaux Platonic solids.
- **Bonds.** How pieces fill space together: exact kits, packings and lattices, from the Euclid–Kepler–Pacioli Cell to the OG on every stacking.
- **Shells.** Pieces closing into nested shells round a core, each layer filled exactly: the Chain of Karma, the Spheres of Dharma, the Bodhi Seed and Theaetetus.
- **Cycles.** Rules that turn one shell into the next and return, with or without end: Universal PiTet, the OG family steps, Kyushindo Cores and the polar reversals.

7. The bridge between the periodic and the aperiodic

Some patterns repeat, like the tiles of a bathroom floor. Others never repeat, yet are perfectly ordered, like the Penrose tiling. These works use the same pieces to build both.

For most of history, order in matter meant repetition. A crystal was a single cell copied without end, and five-fold symmetry was thought impossible in it. In 1982 Dan Shechtman found a metal alloy with sharp five-fold diffraction. Quasicrystals, ordered but never repeating, earned him the Nobel Prize in 2011. Matter has two kinds of order, and the question became how they meet.

The quanta in these works sit on that border. One exact kit builds the repeating octet and also a Fibonacci crystal that never repeats (#40, *The DICTO OG Family on Every Stacking*). In *A Fibonacci Spine* (#30), five repeating packings of the OG meet round a five-fold axis. The copies removed along that axis form exactly the Fibonacci chain, the simplest never-repeating order. The golden shells of the Spheres of Dharma (#23) repeat only by growing φ^3 larger at each turn. One kit, two orders: the pieces themselves are the bridge.

8. What the pieces show

Three things recur. Finite kits are enough: a handful of exact shapes builds shells and spaces without limit. The golden ratio appears wherever five-fold symmetry does, and it decides when shells close. And the pieces join the two orders of matter, the repeating and the never-repeating.

The ancients guessed that space was made of shaped pieces, and modern physics has returned to the idea. These works build such pieces exactly and test them, and each paper that follows ends in a proof.

Sources for the introduction: Plato, Timaeus (tr. B. Jowett); Lucretius, On the Nature of Things (tr. W. E. Leonard); Aristotle, On the Heavens III.8 (tr. J. L. Stocks); J. Kepler, Mysterium Cosmographicum (1596) and Strena seu de nive sexangula (1611); D. Shechtman, I. Blech, D. Gratias, J. W. Cahn, Phys. Rev. Lett. 53 (1984) 1951; C. Rovelli, L. Smolin, Nucl. Phys. B 442 (1995) 593; L. Freidel, S. Speziale, Phys. Rev. D 82 (2010) 084040; E. Bianchi, P. Doná, S. Speziale, Phys. Rev. D 83 (2011) 044035; S. Surya, Living Rev. Relativ. 22 (2019) 5.

Papers (39), in four chapters

Atoms (12)

Single solids, each an exact new piece.

#	Work	About	DOI
#18	**The DICTO Hexa** (2026)	A cube of 8 DICTO Hexa Jewels, its Hexa-Key and their 5 : 3 checkerboard.	DICTO DOI: 10.5281/zenodo.23376320
#19	**AXE, CLEO and FUJI** (2026)	The three pieces that close DICTO's thirteen-icosahedron cluster into one icosahedron of edge φ^2. AXE and CLEO are Mosseri–Sadoc tiles; FUJI is new. With printable nets.	DICTO DOI: 10.5281/zenodo.23351291
#20	**The FUJI Family** (2026)	FUJI as one member of a family of pyramids on flat-ended triangles, indexed by the antiprisms. With the golden breathing pyrochlore, the wheel bound, and Cleo-Icos & Antonius, a new space-filling pair from the golden rhombohedron.	DICTO DOI: 10.5281/zenodo.23357896
#28	**The DICTO Truncated-Icosahedral Hecatodihedron** (2026)	A convex solid of 102 faces from the Spheres of Dharma: the truncated icosahedron with a low pyramid on each hexagon, at the height where neighbouring triangles flatten into rhombi; exactly, the joined truncated icosahedron with its five-fold corners cut flat. Version 2 adds the DICTO Hecatogdoecontahedron.	DICTO DOI: 10.5281/zenodo.23360857
#31	**The DICTO Ogdoecontahedron** (2026)	The Spheres of Dharma turned inside out: the polar dual of the Hecatodihedron, a convex solid of 20 regular hexagons and 60 kites, exact in every coordinate. Joined on its hexagons it forms pyrochlore networks and a body-centred cubic packing filling exactly 77.43% of space.	DICTO DOI: 10.5281/zenodo.23404688
#31a	**The DICTO OCD Stella** (2026)	A short introduction: the icosahedral hull of the golden adapter pieces round one Ogdoecontahedron in its body-centred cubic packing, cut along the 8 joints: 48 corners, 50 faces; copies fill exactly $151/2 - 267\sqrt{5}/8 \approx 87.12\%$ of space, and twelve round a small one fill an outer one exactly at φ^2; with regular octahedra on the second network its crystal fills $1423/32 - 3113\sqrt{5}/160 \approx 96.33\%$.	DICTO DOI: 10.5281/zenodo.23430870
#34a	**The DICTO Hecatogdoecontahedron** (2026)	A short introduction: the DICTO Hecatodihedron and its polar, the DICTO Ogdoecontahedron, interlocked in the truncated icosahedron's midsphere as one self-dual compound, 182 faces and 182 corners; core and hull are the classical rectified and joined truncated icosahedron; the pairing and the name are what is new.	DICTO DOI: 10.5281/zenodo.23430868

#35a	**The DICTO Metaxyhexecontahedron** (2026)	A short introduction: the hull of an icosahedron and a dodecahedron at edge ratio $9/\varphi^4$, in closed form, exact; the solid that closes the golden cycle's last gap, touching the rhombic triacontahedron and the next icosahedron.	DICTO DOI: 10.5281/zenodo.23430864
#35b	**The DICTO Palintriacontahedron** (2026)	A short introduction: the icosidodecahedron's central pyramids flipped out at full height; the hull of the 32 tips, 60 triangles, each half a rhombic-dodecahedron rhombus: the member of the dodecahedron–icosahedron hull family at edge ratio $3/\sqrt{5}$, exact. It interlocks with the next icosahedron, bridging one turn of the golden cycle to the next.	DICTO DOI: 10.5281/zenodo.23430866
#41	**The DICTO Jewel** (2026)	Euclid's dodecahedron carved by its neighbours' stellas: twelve Penrose windows, volume 3/16, and with stellas it fills space; the Jewel-Tetra, Jewel-Octa and Jewel-Rhombi, every tiling with loose stellas, the octet pattern, and the Hexa and Hexa-Key, all exact.	DICTO DOI: 10.5281/zenodo.23496353
#45	**The Reuleaux Platonic Solids** (2026)	DICTO's idea: a ball on every corner of a Platonic solid, as wide as the solid; the common part, named by its look. Proved: the narrowest width is exactly the solid's span, and along each face axis the width is $2\sqrt{(3/4 + r^2)} - 2r$, so dual pairs agree exactly; the face axis as the widest is a conjecture.	DICTO DOI: 10.5281/zenodo.23529499
#49	**The Polyhedral Stretching Theorem** (2026)	DICTO's idea: the long hexagon he saw in the dodecahedron layer of #47, a void with two dodecahedra overlapping it. Proved: slide any solid with equal edges one edge-length, not along one of its edges, and every edge of the swept shape stays the same length; for the dodecahedron this gives the DICTO stretched solids, with three- and four-cap forms, and the icosahedron gives a new-looking family.	DICTO DOI: 10.5281/zenodo.23566671

Bonds (14)

How pieces fill space together: exact kits, packings and lattices.

#	Work	About	DOI
#8	**The Euclid–Kepler–Pacioli Cell** (2026)	All five Platonic solids nest on 13 points in one periodic cell.	DICTO DOI: 10.5281/zenodo.23284106
#13	**The DICTO Dodeca-13** (2026)	The 13-dodecahedron cluster made solid, with two exact gap pieces.	DICTO DOI: 10.5281/zenodo.23284281
#14	**The DICTO-Star** (2026)	An icosidodecahedron, 12 dodecahedra and 20 tridiminished icosahedra closing exactly.	DICTO DOI: 10.5281/zenodo.23284213
#17	**Kagome Hulls** (2026)	Six corner-sharing networks of solid clusters, one rule and its icosahedral exception.	DICTO DOI: 10.5281/zenodo.23376317
#25	**DICTO Spheroid-13 RD** (2026)	A true rhombic dodecahedron wrapped by twelve keyed pieces: a nearly round block of thirteen that fills space by translation alone.	DICTO DOI: 10.5281/zenodo.23332821
#26	**DICTO ATOM-13×13** (2026)	Thirteen clusters of twelve around one, six golden and seven periodic, side by side, with 24 buildable kits.	DICTO DOI: 10.5281/zenodo.23333183
#30	**A Fibonacci Spine** (2026)	Five periodic packings of the DICTO Ogdoecontahedron meet round a five-fold axis; with the central copy, the copies removed along the axis are exactly the canonical Fibonacci chain, checked exactly, until minimal removal thins it at height 12.8.	DICTO DOI: 10.5281/zenodo.23396361
#36	**Golden Bodhi Pyrochlore** (2026)	The gap of the Ogdoecontahedron's bcc packing, exact: 72 cone pieces, limbs crossing in golden rhombi, the canonical golden cut that shatters, the GBPC that every golden fill needs, and no face-to-face tiling of what remains by golden rhombohedra; DICTO's Bull-horns OG adaptor fills the whole gap with one piece, three per copy.	DICTO DOI: 10.5281/zenodo.23451210
#37	**The DICTO Ogdoecontahedron on the Octet Truss** (2026)	Exact space-filling kits with the OG in the octahedra of the octet truss: golden-section keys; an exact Penrose link between facing five-fold stars, with three Penrose cores; golden envelopes filling exactly $10/\varphi^6$ of space, with a two-shape kit and a tetartoid core; and one-piece glue with exact OG imprints, keyed with the OG's own kites so that twelve lock round one OG.	DICTO DOI: 10.5281/zenodo.23457016
#37a	**The DICTO Ogdoecontahedron on the Gyrated Octet** (2026)	The companion to #37: in the gyrated octet the OGs twin across every layer (the OG turned 60°, the spinel law); one exact cell kit for every stacking, with twin caps, bipyramids and a mirror middle piece; the OCD Stella twins too, and the DICTO Belted Triacontahedron fills the octahedra pole to pole; the DICTO Octahedrally Truncated Triacontahedron fits every stacking, and each solid with one fill piece fills space. Version 4: it is the DICTO Jewel's packing, not the solid, that does not twin.	DICTO DOI: 10.5281/zenodo.23484012

#40	**The DICTO OG Family on Every Stacking** (2026)	One exact kit builds the octet, the gyrated octet, the classical polytypes and a Fibonacci crystal that never repeats. Two short rules say why: the twin rule (the mirror twin is the solid turned 60°, the spinel law) and the fit rule (a solid that stays in its layer fits every stacking); the DBT fits every stacking, and the golden envelope fails at twin planes by exactly the tips the DBT removes.	DICTO DOI: 10.5281/zenodo.23490114
#42	**The DICTO Sunstar** (2026)	The Jewel's twin: regular dodecahedra and Dogstars fill space; the Dogstar as an exact stellation, the Star Chain Reaction, the Sunstar-Tetra, Sunstar-Octa and Sunstar-Rhombi, and the Sunstar-Hexa and its Key, all exact.	DICTO DOI: 10.5281/zenodo.23497702
#44	**Aristotle's Pyramid** (2026)	DICTO's question: what could Aristotle have meant by the pyramid that fills space? Proved: of the five Platonic face pyramids, only the cube's alone, the tetrahedron's with the octahedron's, or these together fill space; the dodecahedron's and the Cleo-Icos never do. The golden fix: Cleo-Icos with Antonius; the whole solids with Euclid's roofs.	DICTO DOI: 10.5281/zenodo.23529497
#47	**Every Mix of the Platonic Solids Fills Space** (2026)	DICTO's idea: connect the Platonic solids in any combination. Proved: every one of the 31 mixes, all at one edge length, fills space with only seven helper pieces, among them DICTO's Admiral Caps and the Dogstar.	DICTO DOI: 10.5281/zenodo.23551326

Shells (7)

Pieces closing into nested shells round a core, each layer filled exactly.

#	Work	About	DOI
#22	**The Chain of Karma** (2026)	The regular icosahedron and dodecahedron filled exactly as shells of each other, alternating without end (built and checked to 20,497 pieces). The centres of the Lotus seed's 13 icosahedra match the 600-cell's central vertex and its 12 neighbours, scaled by φ^2; and its surface has the C60 fullerene pattern. Only the icosahedron is ever subdivided; the gap pieces occur at every scale.	DICTO DOI: 10.5281/zenodo.23310541
#23	**Spheres of Dharma** (2026)	Six icosahedral shells, each filled exactly, repeating every φ^3: icosahedron, icosidodecahedron, dodecahedron, fullerene cage, rhombic triacontahedron.	DICTO DOI: 10.5281/zenodo.23310833
#24	**The Bodhi Seed** (2026)	Twelve rhombic triacontahedra and one more inside a triacontahedron φ^2 times larger; the gap fills exactly with three convex piece types, in two ways.	DICTO DOI: 10.5281/zenodo.23311353
#27	**Capturing the In-betweens** (2026)	The in-between stages of the Chain of Karma, Spheres of Dharma and Bodhi Seed as solids: the great stellated dodecahedron, the excavated dodecahedron, the rhombic hexecontahedron, and a further convex solid, the DICTO Truncated-Icosahedral Hecatodihedron.	DICTO DOI: 10.5281/zenodo.23345502
#29	**Theaetetus** (2026)	An icosahedron held exactly by its Cleo-Icos, grown to the DICTO Hecatodihedron in two exact stories: 215 pieces in closed layers, and 102 clean pieces that cut the icosahedron and dodecahedron with the same twenty cones.	DICTO DOI: 10.5281/zenodo.23378070
#34	**Progenitors** (2026)	The DICTO Hecatodihedron and Ogdoecontahedron as a twin pair: on their own they never close golden; set into the Spheres of Dharma in the dodecahedron's place they close it again on φ^9, and the cycle holds the core and hull of two dual compounds.	DICTO DOI: 10.5281/zenodo.23413184
#35	**Vajrayana** (2026)	A companion to Progenitors: the dodecahedron returns inside the golden cycle as the hull of DICTO's Cleo star, filled exactly by 82 pieces of four kinds; the last loose step is closed by a new solid that touches both sides, the DICTO Metaxyhexecontahedron, each of its two layers one kind of piece; the cycle grows to twelve stations.	DICTO DOI: 10.5281/zenodo.23429911

Cycles (6)

Rules that turn one shell into the next and return, with or without end.

#	Work	About	DOI
#32	**Inverted Shells: The Spheres of Dharma Turned Inside Out** (2026)	Polar reciprocation turns the Spheres of Dharma inside out into a new exact golden cycle, every layer filled by one kind of piece. The great stellated dodecahedron locks into its outer gap at exactly one size, a dual pair fits its inner gap in reverse order, and the cycle repeats inward without end.	DICTO DOI: 10.5281/zenodo.23405029
#33	**Double Vision – Both Cycles in One Space** (2026)	The Spheres of Dharma and their inside-out reversal in one space: one sphere of reciprocation toggles which shells the two cycles share, and no sphere gives both. Cut against each other they make one solid of exact convex pieces, 536 of 12 kinds per turn in one setting and 502 of 11 in the other.	DICTO DOI: 10.5281/zenodo.23407107
#38	**Universal PiTet: The DICTO PiTet Proposition** (2026)	From the Cleo star outward in spherical shells: an endless, self-polar cycle of four solids, icosidodecahedron, rhombic triacontahedron, DICTO GTI and DICTO FPD, closing exactly $10/\varphi^4$ larger each turn, with exact shell pieces, Monkey Staff B, five-sided Vajras and hexagonal Cleos. Version 3: new figures, and the rule's prior art (the three-dimensional antipedal construction).	DICTO DOI: 10.5281/zenodo.23463305
#38a	**The DICTO Kepler–Poinsot–UP Cycle** (2026)	The tall companion to Universal PiTet: the great and small stellated dodecahedra in an endless φ^3 cycle, every shell filled exactly by one pattern, in golden reciprocity.	DICTO DOI: 10.5281/zenodo.23464021
#39	**The DICTO OG Family: Four Steps Up and Four Down** (2026)	Universal PiTet's step run from the Ogdoecontahedron, outward and in reverse: eight new icosahedral solids, the DICTO KRE, PHE, TPP and DPHP up and the DICTO TPH, TQH, TQP and TQHD down, all exact; near the OG the shells nest and fill exactly, then both directions break the same way.	DICTO DOI: 10.5281/zenodo.23488944
#43	**DICTO Kyushindo Cores** (2026)	Towers and spires round exact cores in Fibonacci beats: Cleo Towers, Polar Cleo Towers, the OG core of 80 Aktis, the OTT rays and the OG turned out. Main theorem: any icosahedral solid sits inside φ times its foot-point hull, so the golden turned-out tower nests for ever; the polar Hecatodihedron sequence, the ribbons, the new solids TQP-252 to TQP-672 and the tower catalogue.	DICTO DOI: 10.5281/zenodo.23532013

Findings and apps

Work	About	DOI
Kaleidohedra by DICTO (2026)	DICTO's findings (F1 onwards) with the open apps' code, in versioned releases. Findings published there without a separate paper: F1 DICTO skewed rhombic dodecahedron; F5 Regular-hexagon elongated dodecahedron (found independently; known, Grünbaum 2010); F10 DICTO Jewel (EKP windows); F11 The stretched dodecahedron; F12 Star Chain Reaction; F15 DICTO Jewel clusters; F16 Sunstar clusters	DICTO DOI: 10.5281/zenodo.23173809

Version history

- Version 20 added #49, The Polyhedral Stretching Theorem, and #47, Every Mix of the Platonic Solids Fills Space (listed in its Version 2, which rewords its credit line and widens its prior-art search); #49 is listed in its Version 2, which adds its table of library sources; it also says that the numbers #1, #2 … are paper numbers and that findings are numbered F1, F2 …, and lists the Kaleidohedra findings that way.
- Version 19 opens with the essay Quanta of Space, which reads the works as a study of the pieces from which space is built, from the atoms of the ancients to the quanta of modern physics, and arranges the papers in four chapters: Atoms, Bonds, Shells and Cycles.
- Version 18 added #45, The Reuleaux Platonic Solids; #44, Aristotle's Pyramid; #43, DICTO Kyushindo Cores; #42, The DICTO Sunstar; and #41, The DICTO Jewel.
- Version 17 added #40, The DICTO OG Family on Every Stacking, and #39, The DICTO OG Family: Four Steps Up and Four Down, and lists Version 3 of #38 (new figures, and the prior art for its rule) and Version 4 of #37a (it is the DICTO Jewel's packing, not the solid, that does not twin).
- Version 16 listed the Version 3 updates of #37a (the DICTO Octahedrally Truncated Triacontahedron and the DICTO Fill sets) and #31a (the OCD Stella's crystals).
- Version 15 added #37a, The DICTO Ogdoecontahedron on the Gyrated Octet.
- Version 14 added #38a, The DICTO Kepler–Poinsot–UP Cycle, and recorded that from 20 October 2026 every paper listed here is released under CC BY-SA 4.0 (Attribution-ShareAlike): each has a new version with that licence, and earlier versions keep CC BY 4.0.
- Version 13 added #38, Universal PiTet: The DICTO PiTet Proposition.
- Version 12 added #37, The DICTO Ogdoecontahedron on the Octet Truss.
- Version 11 added #36, Golden Bodhi Pyrochlore.
- Version 10 corrected the paper count and named the seven findings published with the app.
- Version 9 added #34a, The DICTO Hecatogdoecontahedron, and #31a, The DICTO OCD Stella (the third and fourth short introductions, one per new solid).
- Version 8 added #35b, The DICTO Palintriacontahedron (the second short introduction).
- Version 7 added #35a, The DICTO Metaxyhexecontahedron (the first short introduction).
- Version 6 added #35, Vajrayana.
- Version 5 added #34, Progenitors.
- Version 4 lists the Version 2 updates of #28, #31, #32 and #33.
- Version 3 added Double Vision (#33).
- Version 2 added the twelve papers published or given their own DOI since Version 1 (14 October 2026).

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